Lectures – 24, 25, 26

DIJKSTRA'S SHORTEST PATH ALGORITHM

G is a weighted, directed graph G = (V,E) with weight function w, which does not allow negative edge weights.

The weight w(p) of a path, the shortest path weight $\delta(u,v)$ for the pair of vertices u and v, and shortest path are defined as before.

The property of optimal sub-structure holds, and so greedy strategy works.

Predecessor subgraph, specified source vertex s and shortest path tree rooted at s is are exactly as defined before.

Relaxation – as before, a key step.

Here d[v] is an attribute of every vertex v in V, which is an upper bound on the weight of the shortest path from s to v. We call it **shortest path estimate**. It is initialized to ∞ for all the vertices except s, for which it is set to 0.

```
RELAX(u,v,w)
  if d[v] > d[u] + w(u,v)
    d[v] = d[u] + w(u,v); // d[v] and pred[v]
    pred[v] = u;         // are updated
```

The predecessor **pred[v]** or π[v] of every vertex v in V is initialized to NULL. Recall the almost identical step in earlier algorithms.

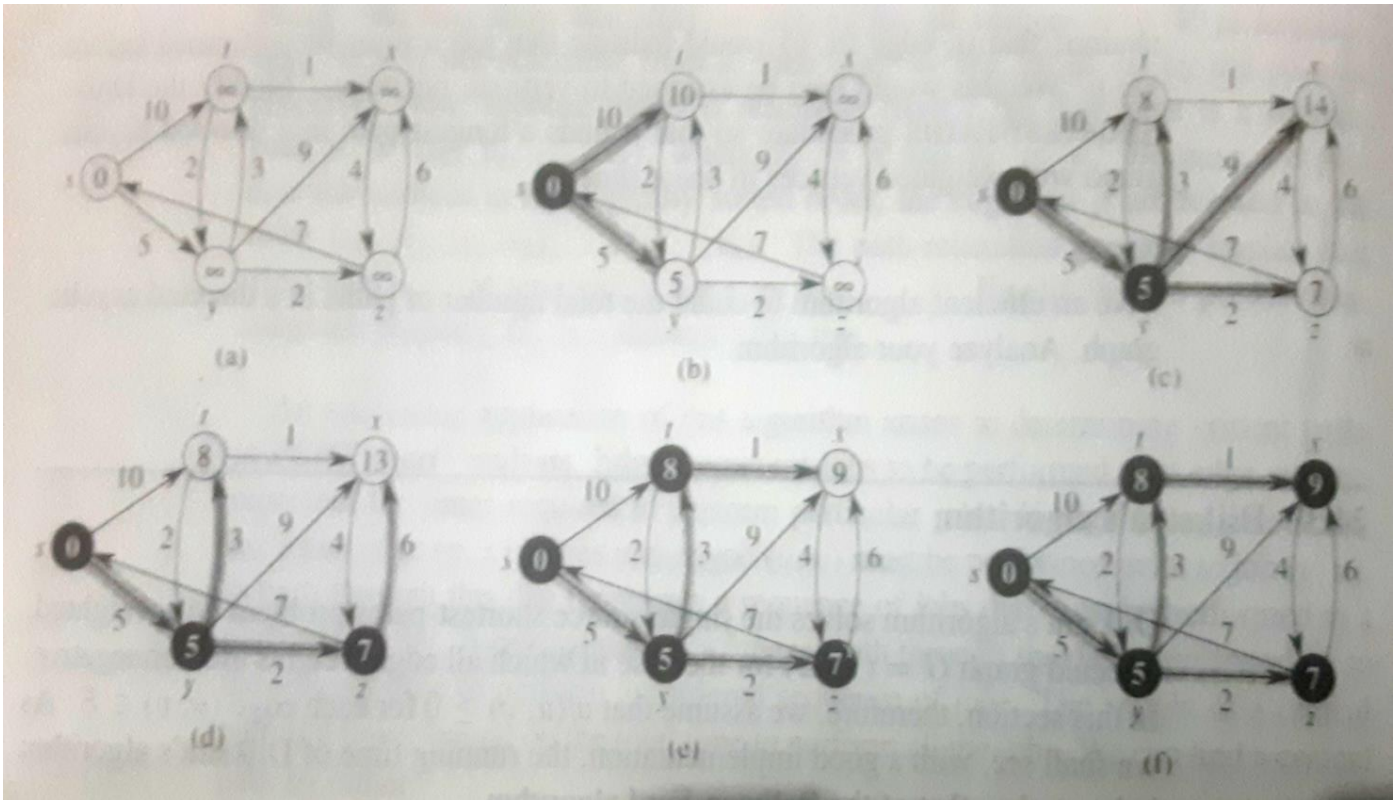
The following is a straightforward function for initialization:

```
INITIALIZE-SINGLE-SOURCE(G,s)
  for each vertex v in V[G]
    d[v] = ∞;
    pred[v] = NIL;
  d[s] = 0;
```

Now we consider Dijkstra's algorithm:

```
DIJKSTRA(G,w,s)
// Initialization
  INITIALIZE-SINGLE-SOURCE(G,s);
  S = NULL;     // S is a set of processed vertices
  Q = V[G];     // Q is a min-priority queue
// Main loop
  while Q <> NULL
    u = EXTRACT-MIN(Q);
    S = S ∪ {u};
    for each v in ADJ[u]
      RELAX(u,v,w);
```

Example [from CLRS]:



Running time:

O(E lgV) if the min-priority queue is implemented using binary min-heap.

O(V lgV + E) if the min-priority queue is implemented using Fibonacci heap.

Question: What is the difference between a “sparse" and a “dense" graph?

ALL-PAIRS SHORTEST PATHS

Common example: Table of road distances between every pair of cities.

Important to note: Road network can be assumed to be ***planar***.

Suppose we run Dijkstra's shortest path algorithm N times, for N nodes, making each node the source node one by one. Of course, for this we must assume non-negative edge weights.

Then the all-pairs shortest paths can be computed in O(VE lgV) time. This is alright if the graph is sparse; improvement is possible with Fibonacci heap.

Matrix methods are useful in solving the all-pairs shortest paths problem. Therefore we shall use the adjacency matrix representation of graphs.

Element w_{ij} in the adjacency matrix W is given by:

$w_{ij} = 0$, if i = j,

$w_{ij} = \infty$, if there is no edge in G from vertex i to vertex j,

w_{ij} = edge weight, if there is an edge in G from vertex i to vertex j.

Note: We assume that there are no negative edge weights.

Predecessor subgraphs

Recall that, when you consider the shortest paths to every other node from a given node i, you get a predecessor subgraph tree rooted at that node. **This tree helps us to re-construct the shortest path.**

So now, considering all N possible root nodes in turn, we will have a total of N predecessor subgraph trees, the i^{th} tree being rooted at node i, for i = 1,2 ... N.

The information about all these N predecessor subgraph trees can be stored in an NxN matrix, compiled while computing the shortest paths.

Some more details are given in [CLRS].

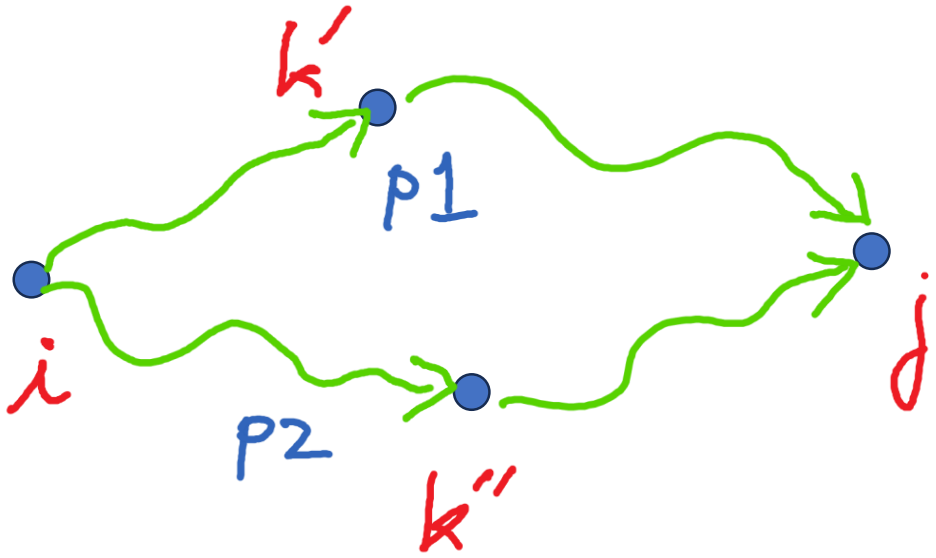
Brief review of usual matrix multiplication

Say n x n matrices A and B are multiplied to give n x n matrix C. That is, C = AB.

$$c_{ij} = \sum_{k=1,n} (a_{ik} * b_{kj})$$

Note the arithmetic operations involved: n multiplications and n additions, in calculating each c_{ij}, i,j = 1, 2, ... n. Assume that multiplications dominate. Then it follows that the straightforward matrix multiplication takes $\Theta(n^3)$ time.

Now think of k as being an intermediate vertex in G on a path from vertex i to vertex j. And consider two such paths.



In the diagram, there are two such paths: p1 and p2. Vertex k' is intermediate on path p1, and vertex k" is intermediate on path p2.

Length of p1 = length of p1<i,k'> **+** length of p1<k',j>

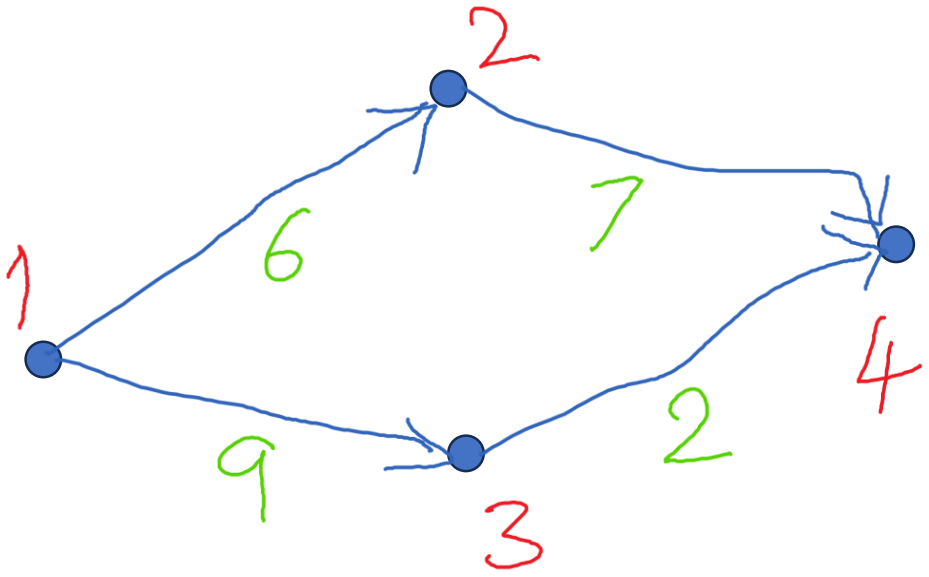
Length of p2 = length of p2<i,k”> **+** length of p2<k”,j>

Shorter path length from i to j = **min** (length of p1, length of p2)

The two key operations here are: **addition** and **minimum**.

It was discovered early on that we can apply the basic concept of matrix multiplication to our problem! The trick is to use **addition** and **minimum** in place of the usual two arithmetic operations.

Simple example – to be solved manually



FLOYD-WARSHALL ALGORITHM FOR ALL-PAIRS SHORTEST PATHS

[Highly simplified from CLRS]

Consider any two vertices i and j in the weighted, directed graph G. Now consider the distance $d_{ij}^{(k)}$ from vertex i to vertex j defined as follows:

$d_{ij}^{(k)}$ = distance of the shortest path from vertex i to vertex j
subject to the condition that **no intermediate vertex on the path is numbered $\geq$ k**

Recursive solution

A recursive solution to the problem is based on the following two observations:

$d_{i,j}^{(0)} = W_{i,j}$ because when k = 0, there are no intermediate vertices on the path from vertex i to vertex j.

$d_{i,j}^{(k)} = \min(d_{i,j}^{(k-1)} , d_{i,k}^{(k-1)} + d_{k,j}^{(k-1)})$ for k = 1, 2, ... n

This is because the shortest path of the type considered, from vertex i to vertex j, **MUST** satisfy one of these two conditions:

(1) it does NOT contain intermediate vertex k, OR

(2) it contains exactly one occurrence of intermediate vertex k.

[This path, by definition, does not contain vertices numbered higher than k.]

Floyd-Warshall algorithm is based on the above two observations.

```
FLOYD-WARSHALL(W)
  n = rows[W];
  D(0) = W;
  for k = 1 to n
    for i = 1 to n
      for j = 1 to n
        D(i,j)(k) = min( D(i,j)(k-1),
                         D(i,k)(k-1) + D(k,j)(k-1) );
  return D(n);
```

Note: Operation of the function **`min(x,y)`** here is very similar to what is done within the function **`RELAX`** which we studied earlier.

The predecessor subgraph**s** are also constructed in a similar way.
[Shown later in CLRS].

Running time analysis: Straightforward. $\Theta(n^3)$.

Example: See below.

First we will draw the graph based on the matrix $W = D^{(0)}$.

$$D^{(0)} = \begin{pmatrix} 0 & 3 & 8 & \infty & -4 \\ \infty & 0 & \infty & 1 & 7 \\ \infty & 4 & 0 & \infty & \infty \\ 2 & \infty & -5 & 0 & \infty \\ \infty & \infty & \infty & 6 & 0 \end{pmatrix} \quad \Pi^{(0)} = \begin{pmatrix} \text{NIL} & 1 & 1 & \text{NIL} & 1 \\ \text{NIL} & \text{NIL} & \text{NIL} & 2 & 2 \\ \text{NIL} & 3 & \text{NIL} & \text{NIL} & \text{NIL} \\ 4 & \text{NIL} & 4 & \text{NIL} & \text{NIL} \\ \text{NIL} & \text{NIL} & \text{NIL} & 5 & \text{NIL} \end{pmatrix}$$

$$D^{(1)} = \begin{pmatrix} 0 & 3 & 8 & \infty & -4 \\ \infty & 0 & \infty & 1 & 7 \\ \infty & 4 & 0 & \infty & \infty \\ 2 & 5 & -5 & 0 & -2 \\ \infty & \infty & \infty & 6 & 0 \end{pmatrix} \quad \Pi^{(1)} = \begin{pmatrix} \text{NIL} & 1 & 1 & \text{NIL} & 1 \\ \text{NIL} & \text{NIL} & \text{NIL} & 2 & 2 \\ \text{NIL} & 3 & \text{NIL} & \text{NIL} & \text{NIL} \\ 4 & 1 & 4 & \text{NIL} & 1 \\ \text{NIL} & \text{NIL} & \text{NIL} & 5 & \text{NIL} \end{pmatrix}$$

$$D^{(2)} = \begin{pmatrix} 0 & 3 & 8 & 4 & -4 \\ \infty & 0 & \infty & 1 & 7 \\ \infty & 4 & 0 & 5 & 11 \\ 2 & 5 & -5 & 0 & -2 \\ \infty & \infty & \infty & 6 & 0 \end{pmatrix} \quad \Pi^{(2)} = \begin{pmatrix} \text{NIL} & 1 & 1 & 2 & 1 \\ \text{NIL} & \text{NIL} & \text{NIL} & 2 & 2 \\ \text{NIL} & 3 & \text{NIL} & 2 & 2 \\ 4 & 1 & 4 & \text{NIL} & 1 \\ \text{NIL} & \text{NIL} & \text{NIL} & 5 & \text{NIL} \end{pmatrix}$$

$$D^{(3)} = \begin{pmatrix} 0 & 3 & 8 & 4 & -4 \\ \infty & 0 & \infty & 1 & 7 \\ \infty & 4 & 0 & 5 & 11 \\ 2 & -1 & -5 & 0 & -2 \\ \infty & \infty & \infty & 6 & 0 \end{pmatrix} \quad \Pi^{(3)} = \begin{pmatrix} \text{NIL} & 1 & 1 & 2 & 1 \\ \text{NIL} & \text{NIL} & \text{NIL} & 2 & 2 \\ \text{NIL} & 3 & \text{NIL} & 2 & 2 \\ 4 & 3 & 4 & \text{NIL} & 1 \\ \text{NIL} & \text{NIL} & \text{NIL} & 5 & \text{NIL} \end{pmatrix}$$